

OpenClaw Integration Analyst Assistant

Overview

This project implements a Personal AI Assistant using OpenClaw and Ollama.

The assistant is designed to help Integration Analysts and API Testers perform common tasks such as:

- API discovery
- Authentication analysis
- Test case generation
- Business rule documentation
- Integration guide generation

The assistant uses multiple agents, reusable skills, local LLMs, and external tools to automate API analysis workflows.

Assignment Requirements Coverage

Requirement	Status
OpenClaw Installation	Complete
Communication Channel	OpenClaw Chat Interface
Three Agents	Complete
Three Skills	Complete
Two External Tools	Complete
Two Model Comparison	Complete
Workflow Demonstration	Complete
Documentation	Complete

Architecture

User Request

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Planning Agent

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API Analysis Agent

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Documentation Agent

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Skills Layer

- API Discovery Skill
- Test Case Generation Skill
- Documentation Generation Skill

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Tools Layer

- Local File System
- Web Search

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LLM Layer

- Llama 3.2 (Ollama)
- Qwen 3 8B (Ollama)

Agents

1. Planning Agent

Responsibilities:

- Understand user requests
- Determine required workflow
- Route tasks to the correct agent

2. API Analysis Agent

Responsibilities:

- Analyze API documentation

- Discover endpoints
- Identify authentication methods
- Extract request and response structures

3. Documentation Agent

Responsibilities:

- Generate integration guides
 - Generate business rules
 - Document error handling scenarios
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Skills

API Discovery Skill

Input:

- Developer portal URL
- Swagger/OpenAPI specification
- Postman collection

Output:

- API inventory
- Authentication summary
- Endpoint catalog

Test Case Generation Skill

Input:

- Endpoint definition

Output:

- Positive test cases
- Negative test cases
- Edge test scenarios

Documentation Generation Skill

Input:

- API specifications
- Sample requests and responses

Output:

- Integration guide
 - Business rules
 - Error handling documentation
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External Tools

Tool 1: Local File System

Used for:

- Reading API specifications
- Reading Postman collections
- Saving generated reports

Tool 2: Web Search

Used for:

- Discovering API documentation
 - Collecting reference information
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Models Evaluated

Llama 3.2

Deployment:

- Local
- Ollama

Advantages:

- Lightweight
- Fast

Qwen 3 8B

Deployment:

- Local
- Ollama

Advantages:

- Better reasoning
 - Better structured output
 - Improved tool usage
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Workflow Demonstration

Example User Request:

"Analyze the Shopify API and generate testing scenarios."

Workflow:

1. Planning Agent receives request.
 2. API Analysis Agent discovers APIs and authentication.
 3. Test Case Generation Skill creates test scenarios.
 4. Documentation Agent generates integration guide.
 5. Response is returned to the user.
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Setup

Clone Repository

```
git clone <repository-url>
```

```
cd openclaw-assignment
```

Create Virtual Environment

```
python -m venv venv
```

```
venv\Scripts\activate
```

Install Dependencies

```
pip install -r requirements.txt
```

Start Ollama

```
ollama serve
```

Verify Models

ollama list

Start OpenClaw

openclaw

Future Improvements

- Telegram Integration
 - Vector Database (ChromaDB)
 - LangFuse Observability
 - Long-Term Memory
 - Human Approval Workflows
 - Multi-Agent Collaboration
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Author

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OpenClaw Assignment Submission